3.2 Forces in action

This section provides knowledge and understanding of the motion of an object when it experiences several forces and also the equilibrium of an object. Learners will also learn how pressure differences give rise to an *upthrust* on an object in a fluid.

There are opportunities to consider contemporary applications of terminal velocity, moments, couples, pressure, and Archimedes principle (HSW6, 7, 9, 11, 12).

Experimental work must play a pivotal role in the acquisition of key concepts and skills (HSW4).

3.2.1 Dynamics

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	net force = mass × acceleration; $F = ma$	Learners will also be expected to recall this equation. *M1.1*
(b)	the newton as the unit of force	
(c)	weight of an object; $W = mg$	Learners will also be expected to recall this equation.
(d)	the terms tension, normal contact force, upthrust and friction	
(e)	free-body diagrams	
(f)	one- and two-dimensional motion under constant force.	